DID YOU GET WHAT YOU WANTED? PERSPECTIVES ON FAIRNESS IN EDUCATION ACROSS EUROPE

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Key Takeaways

- Perceived chances to attain the education one wants vary based on **individual (household) characteristics**
- Policy-Makers should not just focus on aspects directly related to education, but especially on those **related to the household**
- **Structural variables** like the youth unemployment seem to have **no strong association**

Background & Motivation

- **Discrepancy** between 'objective' education quality and perceived quality and fairness (Boyadjieva et al., 2024)
- **Regional variables** also influence changes and might moderate associations (Boyadjieva et al., 2024)
- Access to (higher) education researched mostly with **individual characteristics** like gender, race, and income (Naim, 2025)
- Data Summary on Make it Strong indicates differences with respect to **age and educational background** (Stoilova et al., 2025)
- To what extent do **people's perceived chances of attaining the education they wanted** depend on their **own background** and on **structural aspects** of their region?

Methods and Data

- **Linear Mixed Models** with **three levels**:
  - Country (fixed effect)
  - Region (random effect)
  - Individual
- **robustness analysis**, **diagnostics** and variable selection using the **BIC** applied

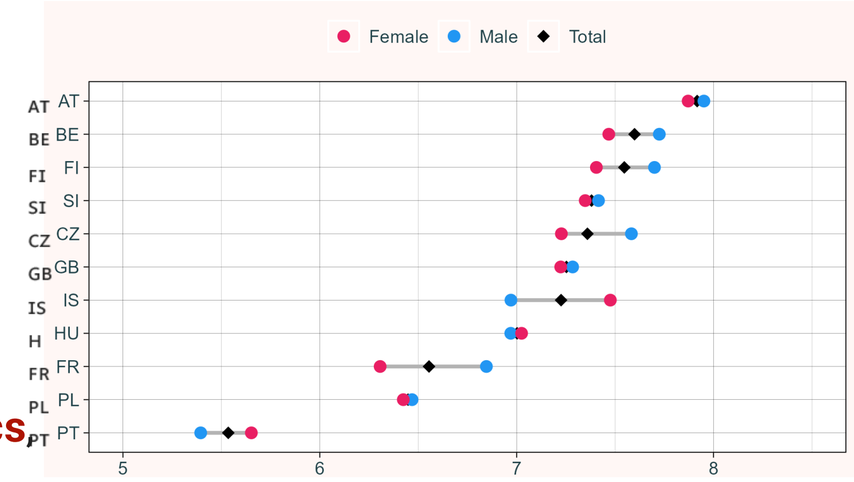

Figure 1: mean perceived fairness by gender (own calculations)
Note: x-axis adjusted for better visibility

Dataset	Pillar / Dataset	Items
CRONOS-3 (Wave 1) (ESS ERIC, 2025a)	Make it **Strong**	- Perceived fairness of own education - Satisfaction with own education
CRONOS-3 (Wave 3) (ESS ERIC, 2025b)	Make it **Strong**	- Current occupation matches education
CRONOS-3 (Wave 5) (ESS ERIC, 2025c)	Make it **Healthy**	- Felt safe with at least one carer in first 18 years - Serious conflict in household in first 18 years - Severe financial difficulties in family first 18 years
Eurostat (2026a)	`ilc_li41`	- Regional At-risk-of-poverty rate
Eurostat (2026b)	`edat_lfse_16`	- Regional early leavers from education
Eurostat (2026c)	`edat_lfse_22`	- Regional young NEET rate

Results

- Variables concerning the **family** (i.e., feeling safe, conflicts and financial difficulties) in youth **significantly associated** for perceived fairness of education
- Results also emphasize existing association with **current income**
- Close association with **the own education**: The more I am satisfied with my own education, the fairer I perceive my chances
- **Female respondents** tend to perceive educational chances as **less fair**
- **Regional attributes** have **almost no predictive power**

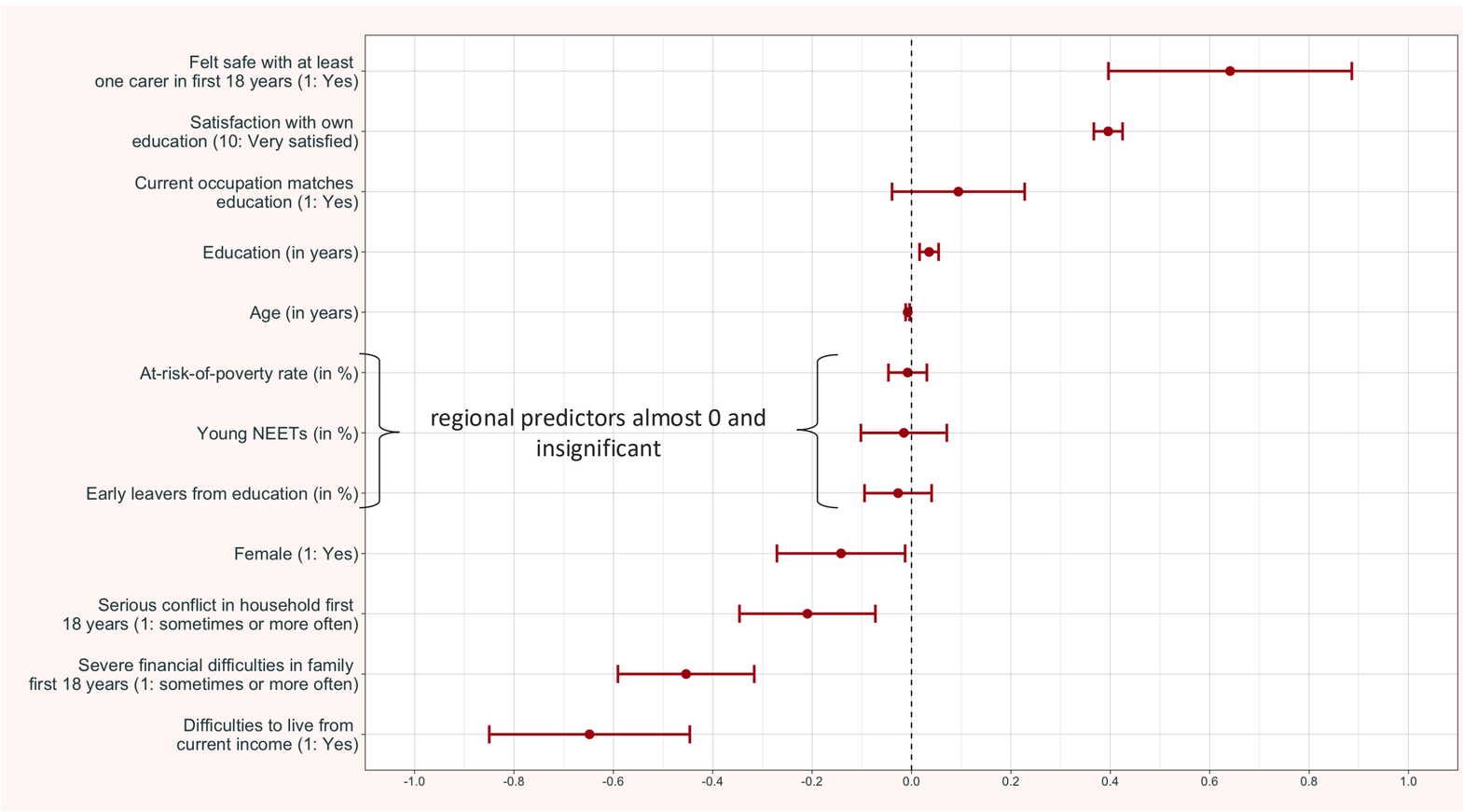

Figure 2: Regression coefficients of linear mixed model. Error bars indicate 95% confidence interval. `w1pspweight` applied.
Note: In the model, fixed effects for countries (dummy variables) are included as well but not depicted here (own calculations).

Discussion

- Findings **do not** suggest strong influence **regional variables**
- **Individual aspects** seem to be more important, especially concerning the childhood household
- **Gender** and **income** remain crucial factors
- Results must be interpreted in context of some **limitations**:
  - missing data on regional level
  - small sample sizes in some regions
- Further research: **small area methods** (e.g., Fay-Harriot models) and focus on **educational quality** in regions

References

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GitHub Repository

https://github.com/nippisch/datathon_make_it_strong

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